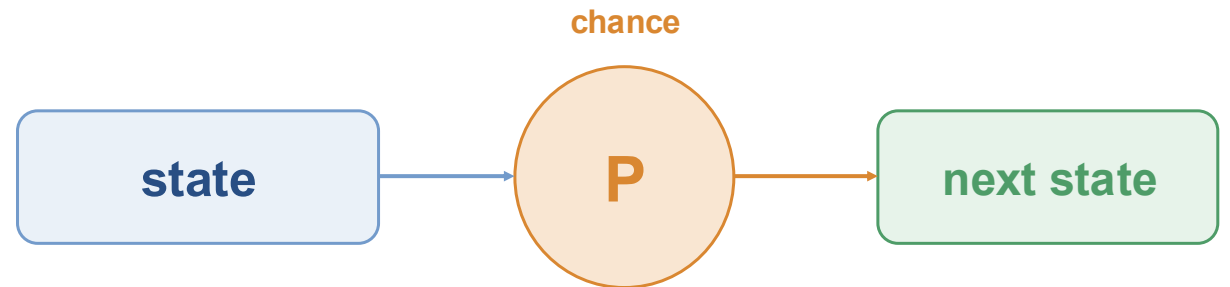


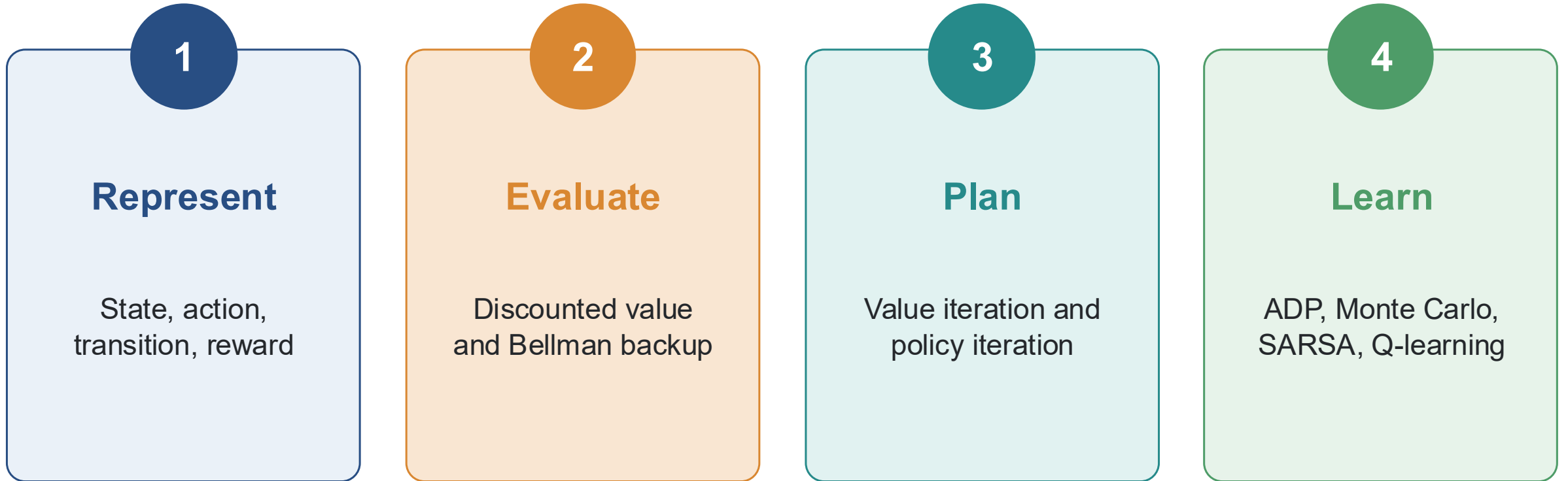


Tutorial 3

Sequential Decision Making under Uncertainty



- The tutorial follows one repeated decision loop



Every question slide contains the information needed to work from the screen.



01 MDP review

The minimum intuition and notation needed for both tutorial problems.

QUESTION DISCUSSION REVEAL

■ A policy reacts after every uncertain outcome

Lecture pp. 2-7, 17

Fixed action sequence

eco, boost, boost, eco, ...

The plan cannot react when chance produces an unexpected state.

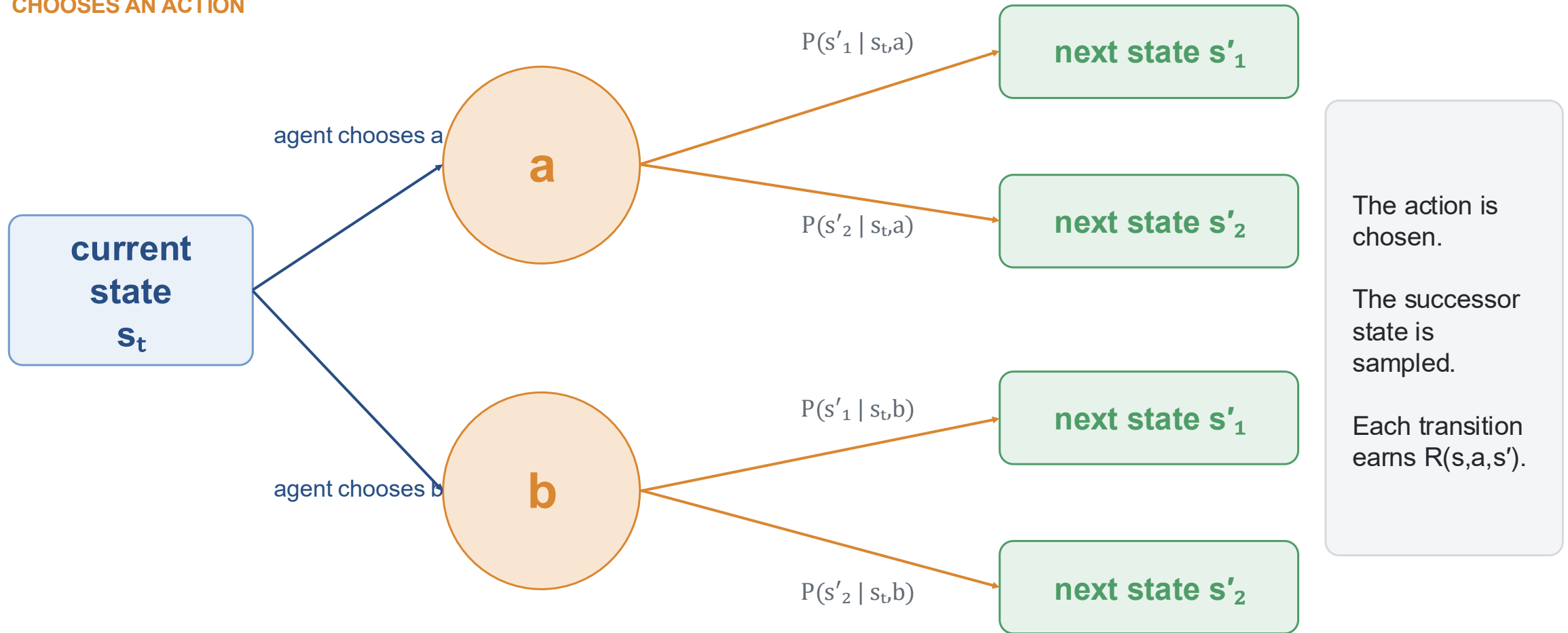
Feedback policy

$$\begin{aligned}\pi(\text{charged}) &= a_C \\ \pi(\text{low}) &= a_L\end{aligned}$$

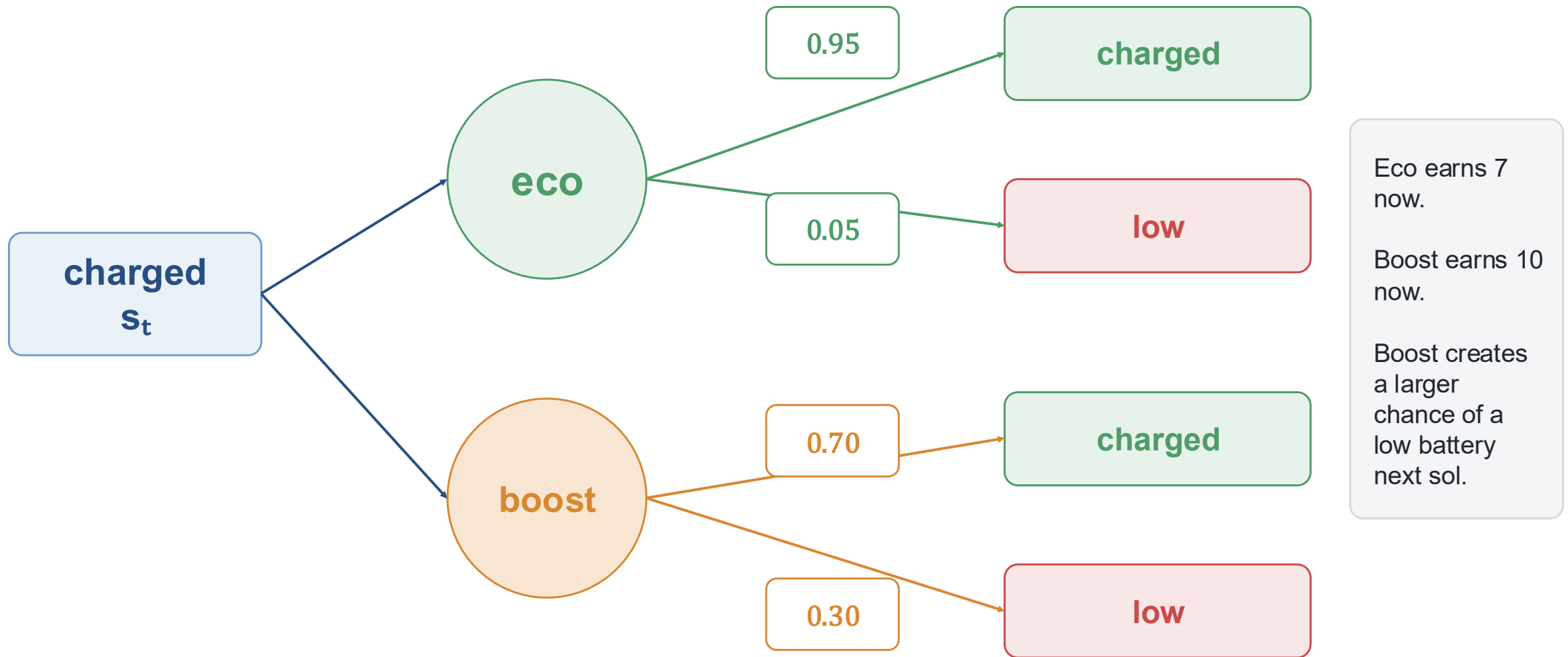
Observe the new state, then choose again.

A chosen action opens a probability tree

CIRCLES = CHANCE STEP AFTER THE AGENT
CHOOSES AN ACTION



- Nova's charged state has two action-conditioned chance nodes



Nova transition and reward data: current Tutorial 3, pp. 1-2.

A Markov Decision Process separates choice from uncertainty

Lecture pp. 10, 13-14

$$M = (S, A, P, R, \gamma)$$

Symbol	Meaning	Nova example
S	states that summarize what matters	charged, low
A	actions the agent may choose	eco, boost
$P(s' s,a)$	distribution over possible next states	$P(\text{charged} \text{charged},\text{eco})=0.95$
$R(s,a,s')$	immediate transition reward	$R(\text{charged},\text{eco})=7$
γ	weight on rewards one step further away	0.8 in Problem 1 Part 2

$$\sum_{s'} P(s'|s,a)=1$$

$$P(S_{t+1}|\text{history},S_t,A_t)=P(S_{t+1}|S_t,A_t)$$

Reward conventions must stay consistent

general: $R(s,a,s')$

Nova: reward depends on s and a

$R(s,a)$

Independent of the successor s' . Values are 7, 10, 0, and 2.

Aster: reward labels the transition

$R(s,a,s')$

Examples: $b \rightarrow T$ gives 6; $b \rightarrow s_1$ gives -2.

Terminal states have value 0 in episodic RL; the transition into a terminal state carries the terminal reward.

- Discounting sets how far ahead the agent cares

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \gamma^3 r_{t+4} + \dots$$

next reward r_{t+1}
weight 1

reward r_{t+2}
weight γ

reward r_{t+3}
weight γ^2

reward r_{t+4}
weight γ^3

Smaller γ
Immediate reward dominates
sooner

Larger γ
Battery health far in the future
matters more

- Value asks what happens now and what the state enables later

$$Q^*(s,a) = \sum_{s'} P(s'|s,a)[R(s,a,s') + \gamma V^*(s')]$$

$$V^*(s) = \max_a Q^*(s,a)$$

$$\pi^*(s) = \arg \max_a Q^*(s,a)$$

$Q^*(s,a)$
fix the first action

$V^*(s)$
best achievable
value

$\pi^*(s)$
action attaining the
max

Nova uses $R(s,a)$, so $\sum_{s'} P(s'|s,a)R(s,a,s') = R(s,a)$. For a fixed-policy value $V^{\pi}(s)$, the max disappears because π specifies the action.

Value iteration and policy iteration use different inner loops

Value iteration

$$V_{k+1}(s) = \max_a \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma V_k(s')]$$

One Bellman optimality backup per sweep. Repeat until values stabilize.

Policy iteration

- 1 evaluate fixed π exactly
- 2 improve π greedily

Evaluation has no max. Improvement compares all actions.

Both converge to the same optimal policy for this discounted finite MDP.

■ Review checkpoint

60 SECONDS, NO NOTES

1 Who chooses the action?

2 What does the chance node sample?

3 Why can a policy outperform a fixed action sequence?

4 Where does the max appear in a Bellman backup?

5 What changes when γ becomes smaller?

6 Which step of policy iteration has no max?

■ Review worksheet

STUDENT WORKING

1

2

3

4

5

6

Recap

tic environment. Whereas Chapter 16 was concerned with one-shot or episodic decision problems, in which the utility of each action's outcome was well known, we are concerned here with **sequential decision problems**, in which the agent's utility depends on a sequence of decisions. Sequential decision problems incorporate utilities, uncertainty, and sensing, and include search and planning problems as special cases. Section 17.1 explains how sequential

Recap: MDP

- An MDP $M \triangleq (S, A, T, R, \gamma)$ consists of:
- A set S of states
- A set A of actions

Recap: MDP

As in Chapter 3, the **transition model** (or just “model,” when the meaning is clear) describes the outcome of each action in each state. Here, the outcome is stochastic, so we write $P(s' | s, a)$ for the probability of reaching state s' if action a is done in state s . (Some authors write $T(s, a, s')$ for the transition model.) We will assume that transitions are **Markovian**: the probability of reaching s' from s depends only on s and not on the history of earlier states.

$$P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = P(s_{t+1} | s_t, a_t)$$

Later in this section, we investigate the nature of utility functions on histories; for now, we simply stipulate that for every transition from s to s' via action a , the agent receives a **reward** $R(s, a, s')$. The rewards may be positive or negative, but they are bounded by $\pm R_{\max}$.¹

- **Formally:**

- Define $R(s, a, s')$ for all $s, s' \in S$ and for all $a \in A$.

- **Alternate forms:** (given $R(s, a, s')$ above)

- $R(s, a)$ as $\sum_{s'} P(s' | s, a) R(s, a, s')$ independent of s'
- $R(s)$ as $\sum_{s'} P(s' | s, a) R(s, a, s')$, independent of a and s'

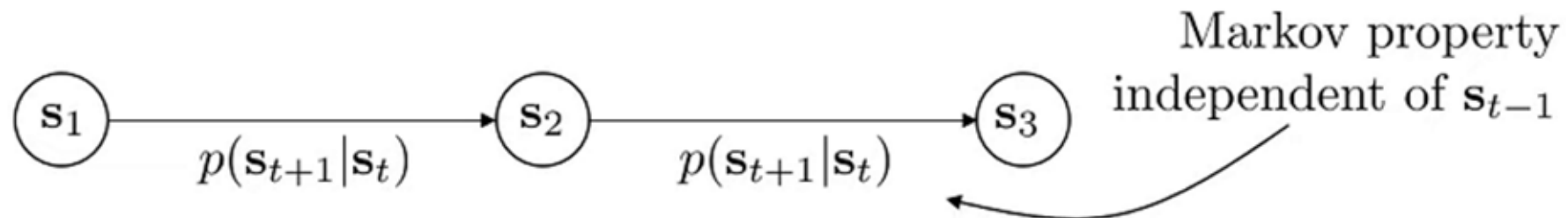
Other possible ways to
define reward functions

goal. Therefore, a solution must specify what the agent should do for *any* state that the agent might reach. A solution of this kind is called a **policy**. It is traditional to denote a policy by π , and $\pi(s)$ is the action recommended by the policy π for state s . No matter what the outcome

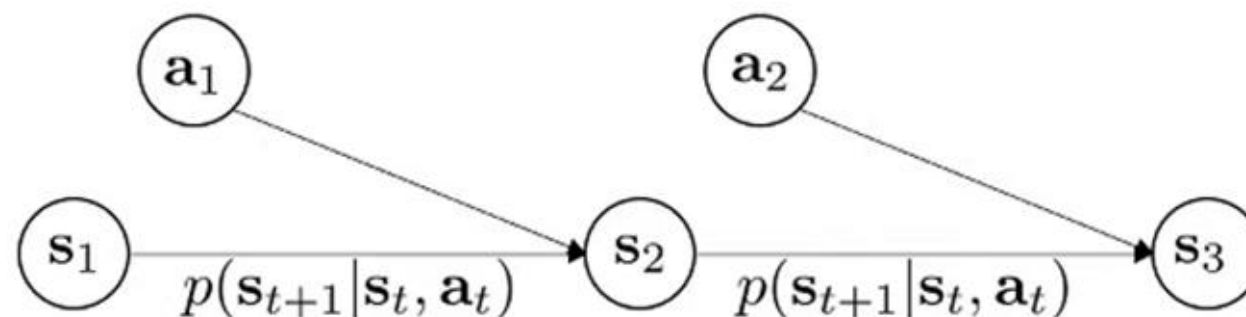
Recap: Markov Property

$$P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = P(s_{t+1} | s_t, a_t)$$

Markov Chain



**Markov
Decision
Process**



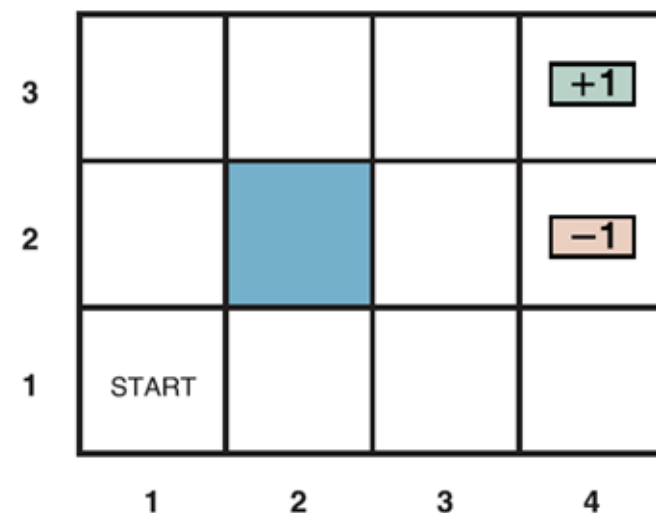
Recap: Finite v.s. Infinite Horizon Utilities

The first question to answer is whether there is a **finite horizon** or an **infinite horizon** for decision making. A finite horizon means that there is a *fixed* time N after which nothing matters—the game is over, so to speak. Thus,

$$U_h([s_0, a_0, s_1, a_1, \dots, s_{N+k}]) = U_h([s_0, a_0, s_1, a_1, \dots, s_N])$$

for all $k > 0$. For example, suppose an agent starts at (3,1) in the 4×3 world of Figure 17.1, and suppose that $N=3$. Then, to have any chance of reaching the +1 state, the agent must head directly for it, and the optimal action is to go *Up*. On the other hand, if $N=100$, then there is plenty of time to take the safe route by going *Left*. *So, with a finite horizon, an optimal action in a given state may depend on how much time is left.* A policy that depends on the time is called **nonstationary**.

With no fixed time limit, on the other hand, there is no reason to behave differently in the same state at different times. Hence, an *optimal action depends only on the current state, and the optimal policy is stationary*. Policies for the infinite-horizon case are therefore simpler than those for the finite-horizon case, and *we deal mainly with the infinite-horizon case in this chapter.* (We will see later that for partially observable environments, the infinite-horizon



Recap: Utilities

assume the agent is in some initial state s and define S_t (a random variable) to be the state the agent reaches at time t when executing a particular policy π . (Obviously, $S_0 = s$, the state the agent is in now.) The probability distribution over state sequences $S_1, S_2, \dots$, is determined by the initial state s , the policy π , and the transition model for the environment.

The expected utility obtained by executing π starting in s is given by

$$U^\pi(s) = E \left[\sum_{t=0}^{\infty} \gamma^t R(S_t, \pi(S_t), S_{t+1}) \right], \quad (17.2)$$

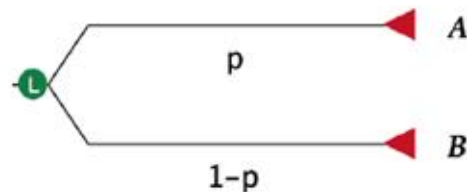
$$U_h([s_0, a_0, s_1, a_1, s_2, \dots]) = R(s_0, a_0, s_1) + \gamma R(s_1, a_1, s_2) + \gamma^2 R(s_2, a_2, s_3) + \dots,$$

• Utility of a Lottery:

- The utility of a lottery is the weighted average of the utilities of each possible outcome:

$$U([p_i, S_i; \dots; p_n, S_n]) = \sum_i p_i U(S_i)$$

- Once the probabilities and utilities of the outcomes are known, the utility of the entire lottery is fully determined.



$$U(L) = pU(A) + (1-p)U(B)$$

Recap: Utilities

assume the agent is in some initial state s and define S_t (a random variable) to be the state the agent reaches at time t when executing a particular policy π . (Obviously, $S_0 = s$, the state the agent is in now.) The probability distribution over state sequences $S_1, S_2, \dots$, is determined by the initial state s , the policy π , and the transition model for the environment.

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$$U^\pi(s) = E \left[\sum_{t=0}^{\infty} \gamma^t R(S_t, \pi(S_t), S_{t+1}) \right], \quad (17.2)$$

of a state and the utility of its neighbors: *the utility of a state is the expected reward for the next transition plus the discounted utility of the next state, assuming that the agent chooses the optimal action.* That is, the utility of a state is given by

$$U(s) = \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')]. \quad (17.5)$$

This is called the **Bellman equation**, after Richard Bellman (1957). The utilities of the

The utility function $U(s)$ allows the agent to select actions by using the principle of maximum expected utility from Chapter 16—that is, choose the action that maximizes the reward for the next step plus the expected discounted utility of the subsequent state:

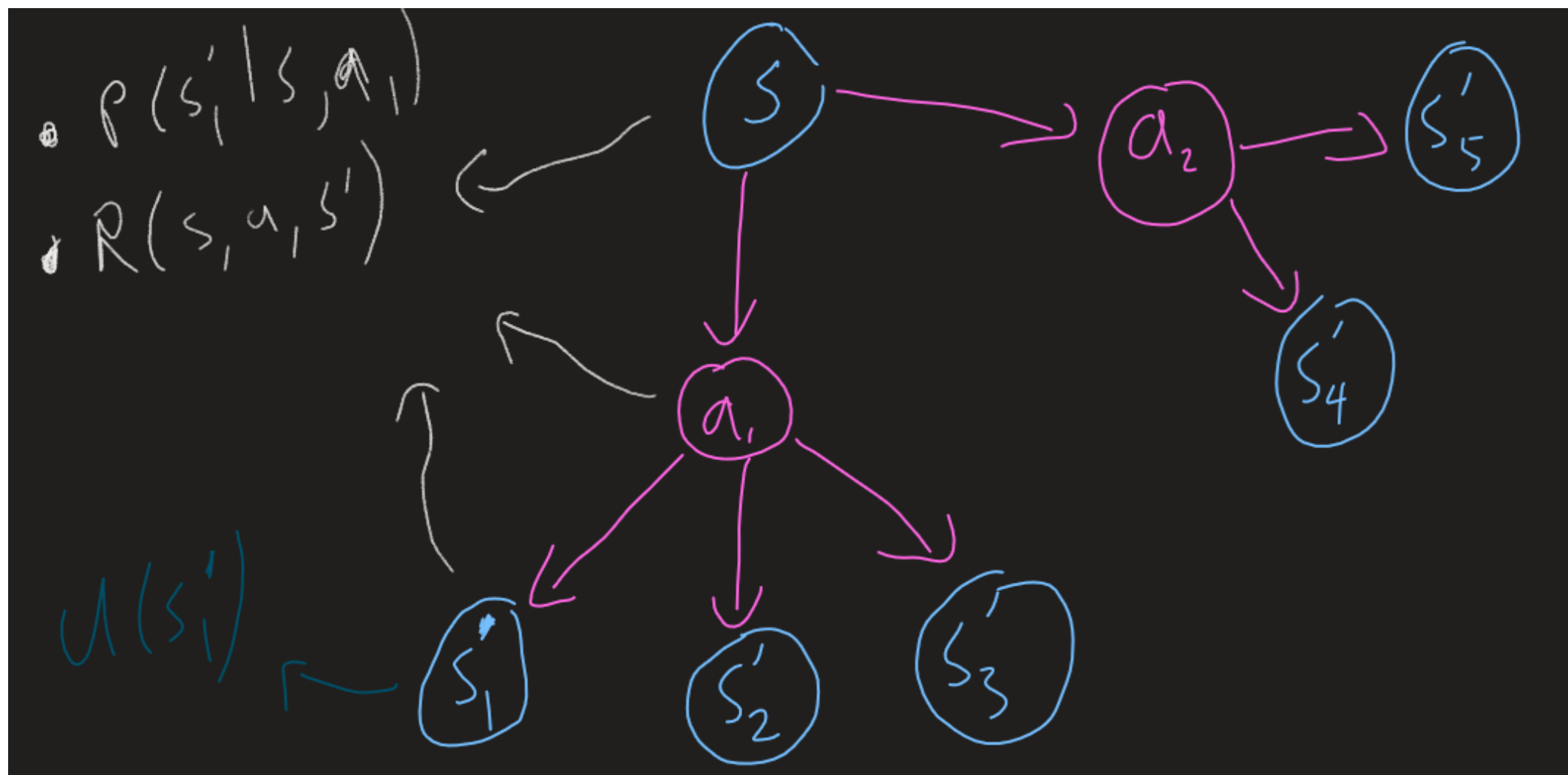
$$\pi^*(s) = \operatorname{argmax}_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')]. \quad (17.4)$$

Recap: Utilities

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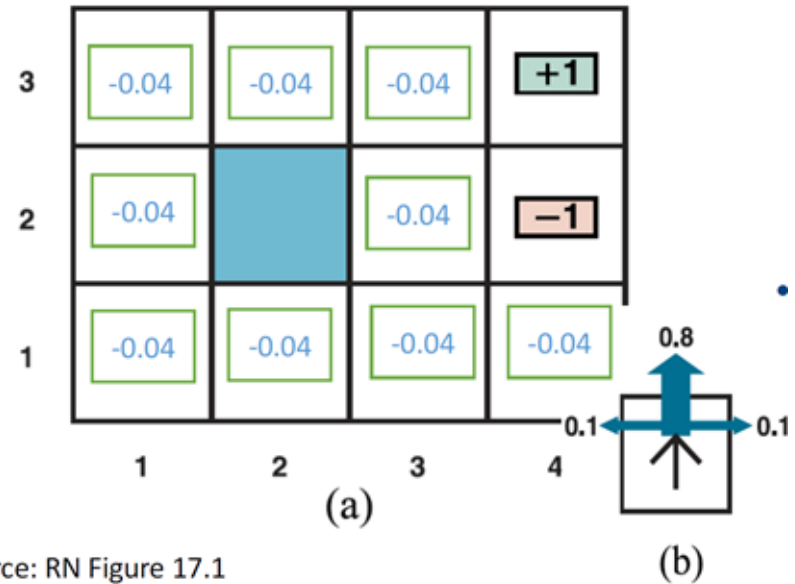
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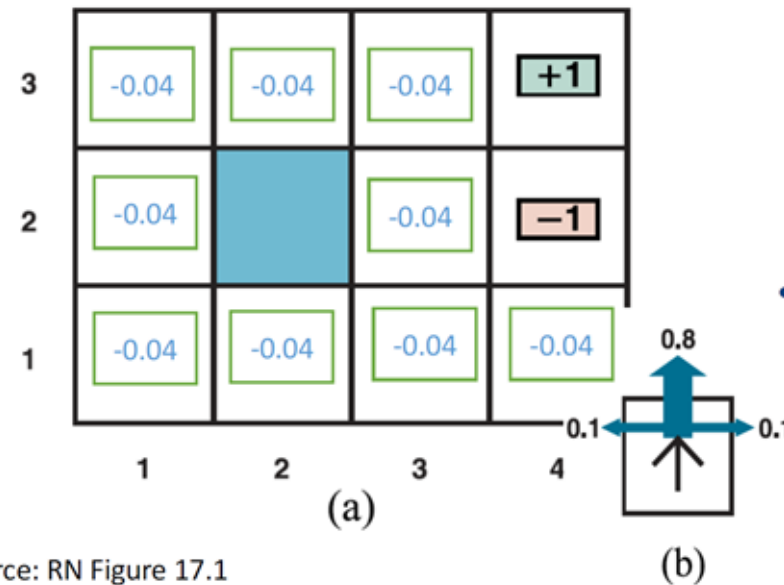


Source: RN Figure 17.1

(U)
(L)
(D)
(R)

Recap: Utilities

$$U(s) = \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')]. \quad (17.5)$$



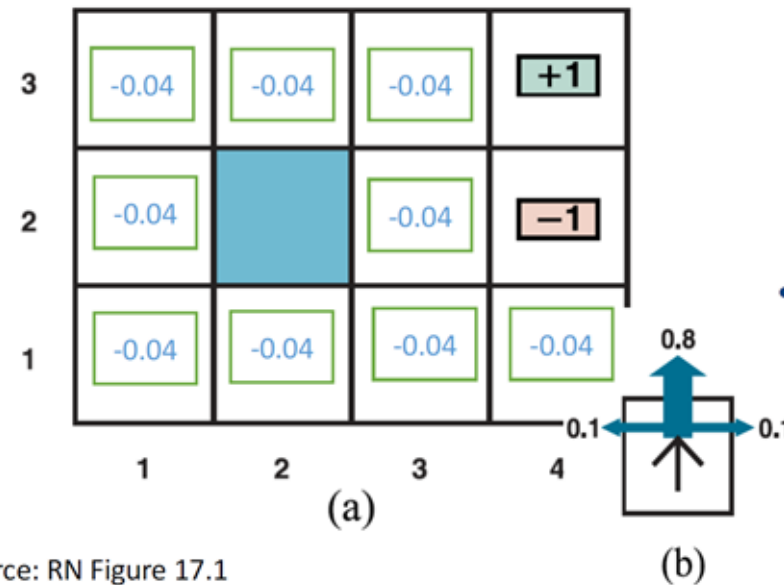
Source: RN Figure 17.1

Let us look at one of the Bellman equations for the 4×3 world. The expression for $U(1,1)$ is

$$\max \{ [0.8(-0.04 + \gamma U(1,2)) + 0.1(-0.04 + \gamma U(2,1)) + 0.1(-0.04 + \gamma U(1,1))], \begin{matrix} (U) \\ (L) \\ (D) \\ (R) \end{matrix} \}$$

Recap: Utilities

$$U(s) = \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')]. \quad (17.5)$$



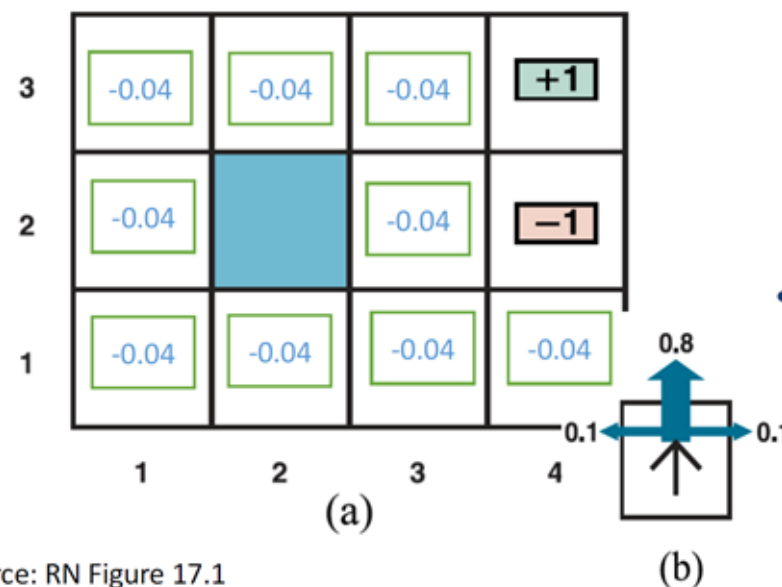
Source: RN Figure 17.1

Let us look at one of the Bellman equations for the 4×3 world. The expression for $U(1, 1)$ is

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Recap: Utilities

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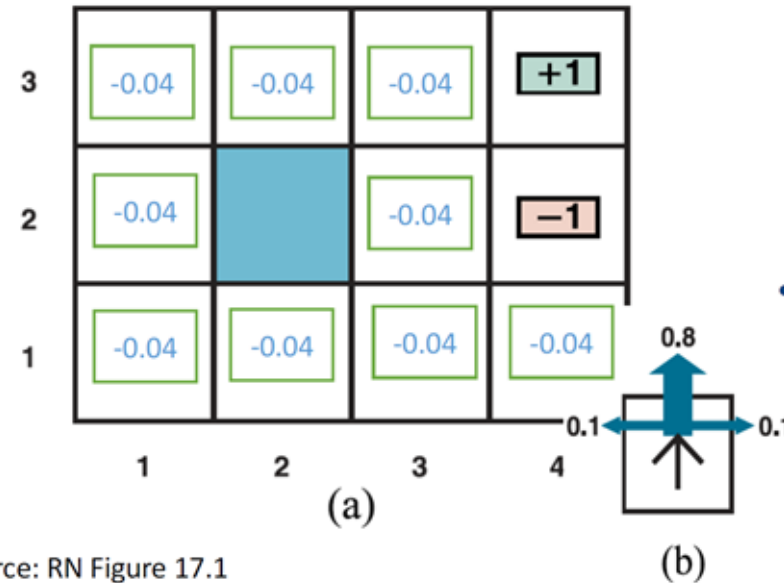
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Recap: Utilities

$$U(s) = \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')]. \quad (17.5)$$



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Recap: Action-Utility Function; Q Function

$$U(s) = \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')].$$

$$\begin{aligned} Q(s, a) &= \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')] \\ &= \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma \max_{a'} Q(s', a')] \end{aligned}$$

Another important quantity is the **action-utility function**, or **Q-function**: $Q(s, a)$ is the expected utility of taking a given action in a given state. The Q-function is related to utilities in the obvious way:

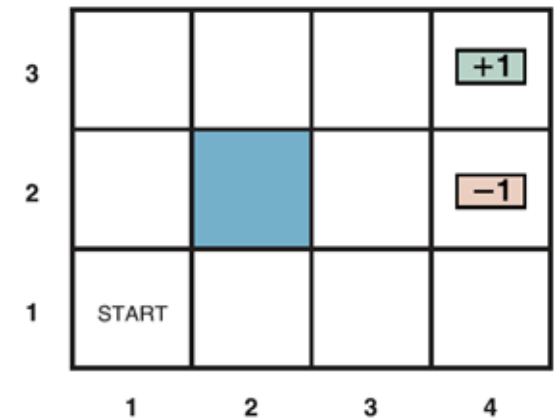
$$U(s) = \max_a Q(s, a). \quad (17.6)$$

Furthermore, the **optimal policy** can be extracted from the Q-function as follows:

$$\pi^*(s) = \operatorname{argmax}_a Q(s, a). \quad (17.7)$$

Recap: Value Iteration

$$U_{i+1}(s) \leftarrow \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U_i(s')],$$



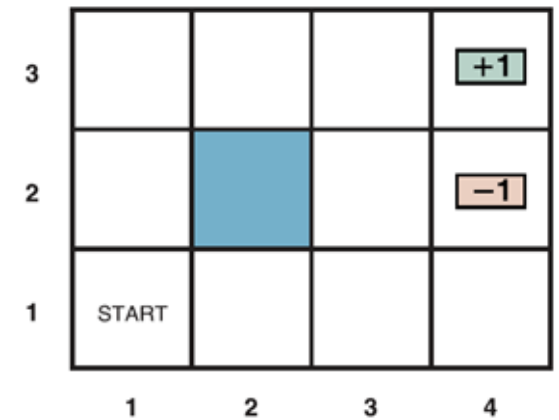
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where the four expressions correspond to *Up*, *Left*, *Down* and *Right* moves. When we plug in

Recap: Value Iteration

$$U_{i+1}(s) \leftarrow \max_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U_i(s')],$$



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where the four expressions correspond to *Up*, *Left*, *Down* and *Right* moves. When we plug in

state. The n equations contain n unknowns—the utilities of the states. So we would like to solve these simultaneous equations to find the utilities. There is one problem: the equations are *nonlinear*, because the “max” operator is not a linear operator. Whereas systems of linear equations can be solved quickly using linear algebra techniques, systems of nonlinear equations are more problematic. One thing to try is an *iterative* approach. We start with arbitrary initial values for the utilities, calculate the right-hand side of the equation, and plug it into the left-hand side—thereby updating the utility of each state from the utilities of its neighbors. We repeat this until we reach an equilibrium.

Recap: Policy Iteration

- **Policy evaluation:** given a policy π_i , calculate $U_i = U^{\pi_i}$, the utility of each state if π_i were to be executed.

$$U_i(s) = \sum_{s'} P(s' | s, \pi_i(s)) [R(s, \pi_i(s), s') + \gamma U_i(s')].$$

Recap: Policy Iteration

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$$U_i(s) = \sum_{s'} P(s' | s, \pi_i(s)) [R(s, \pi_i(s), s') + \gamma U_i(s')].$$

For example, suppose π_i is the policy shown in Figure 17.2(a). Then we have $\pi_i(1,1) = Up$, $\pi_i(1,2) = Up$, and so on, and the simplified Bellman equations are

$$U_i(1,1) = 0.8[-0.04 + U_i(1,2)] + 0.1[-0.04 + U_i(2,1) + 0.1[-0.04 + U_i(1,1)]],$$

$$U_i(1,2) = 0.8[-0.04 + U_i(1,3)] + 0.2[-0.04 + U_i(1,2)],$$

and so on for all the states. The important point is that these equations are *linear*, because the “max” operator has been removed. For n states, we have n linear equations with n unknowns, which can be solved exactly in time $O(n^3)$ by standard linear algebra methods. If the

Recap: Policy Iteration

- **Policy evaluation:** given a policy π_i , calculate $U_i = U^{\pi_i}$, the utility of each state if π_i were to be executed.

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- **Policy improvement:** Calculate a new MEU policy π_{i+1} , using one-step look-ahead based on U_i (as in Equation (17.4)).

$$\pi^*(s) = \operatorname{argmax}_{a \in A(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma U(s')].$$

Recap: Policy Iteration

- **Policy evaluation:** given a policy π_i , calculate $U_i = U^{\pi_i}$, the utility of each state if π_i were to be executed.

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TERMINATE WHEN
POLICY IS THE
SAME



02 Problem 1

Nova the Mars Rover: discounting, policy values, value iteration, and policy iteration.

[QUESTION](#) [DISCUSSION](#) [REVEAL](#)

■ Problem 1: complete setup

OFFICIAL SETUP

Nova collects scientific data while managing a limited battery. Each sol it chooses eco or boost. Boost earns more immediately, but makes a low battery more likely.

$S = \{\text{charged, low}\}$ $A = \{\text{eco, boost}\}$

Immediate reward
boost > eco in both states

Future opportunity
charged states earn much more

Infinite horizon, start in charged state, and use one reward convention consistently.

■ Nova's model and four stationary policies

Transition probability $P(\text{charged} \mid s,a)$

State	Action	$P(\text{charged} \mid s,a)$
charged	eco	0.95
charged	boost	0.70
low	eco	0.50
low	boost	0.10

Reward $R(s,a)$, independent of s'

State	Action	Reward
charged	eco	7
charged	boost	10
low	eco	0
low	boost	2

EE

BB

EB

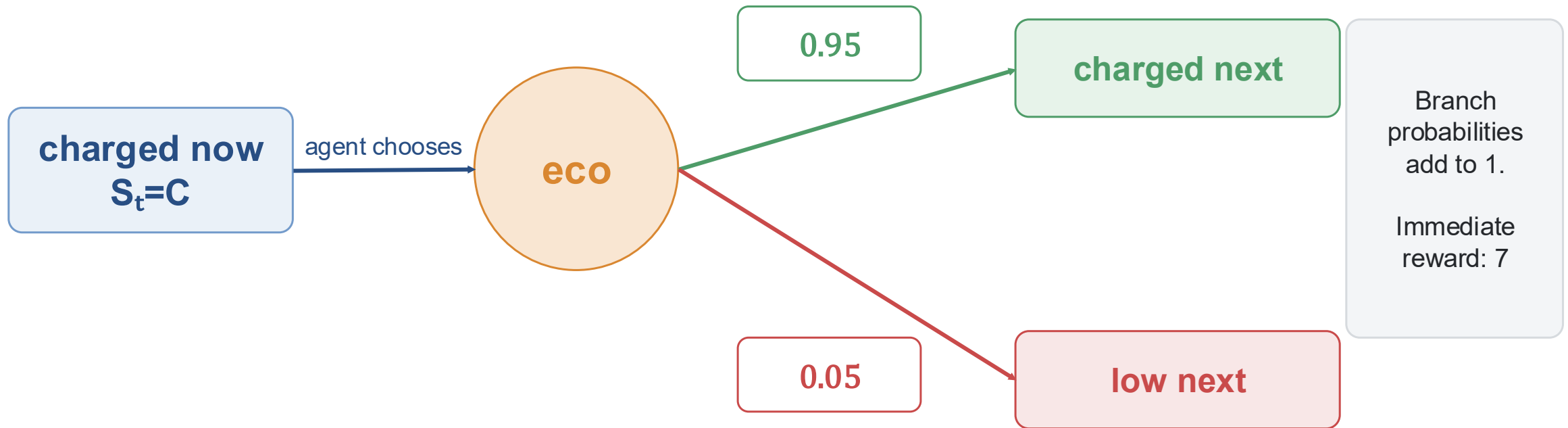
BE

E=eco; B=boost. First letter: charged action; second: low action. Example BE means boost when charged, eco when low.

For every row, $P(\text{low next} \mid s,a) = 1 - P(\text{charged next} \mid s,a)$.

■ Read each transition as: now, action, then chance

Example: Nova is charged now and chooses eco. The model says $P(\text{charged next} \mid \text{charged now, eco})=0.95$.



■ Part 1(a): discounting question

OFFICIAL QUESTION

Start charged. E=eco, B=boost; policy XY uses X when charged and Y when low.

Current state	Action	P(charged next)	Immediate reward
charged	eco	0.95	7
charged	boost	0.70	10
low	eco	0.50	0
low	boost	0.10	2

Explain the role of discounting future rewards in MDPs. How would behavior differ if γ were 0.6 instead of 0.9?

Predict first: which γ should care more about preserving the battery?

Question: current Tutorial 3, p. 2.

■ Compare discount weights

STUDENT WORKING

weight k steps ahead = γ^k

Reward delay	$\gamma=0.6$	$\gamma=0.9$
1 step	0.600	0.900
2 steps	0.360	0.810
5 steps	0.078	0.590
10 steps	0.006	0.349

$\gamma=0.6$
What behavior do you expect?

$\gamma=0.9$
What behavior do you expect?

For bounded rewards and $\gamma < 1$, the infinite discounted return is finite.

■ p_t tracks the probability that Nova is charged

Start condition: S_0 =charged with certainty. Therefore $p_0=P(S_0=\text{charged})=1$ and $P(S_0=\text{low})=0$.

$$p_t = P(S_t=\text{charged}) \quad \text{and} \quad 1-p_t = P(S_t=\text{low})$$

Time	Meaning of p_t	Meaning of $1-p_t$
$t=0$	probability charged now = 1	probability low now = 0
$t=1$	probability charged after one transition	probability low after one transition
$t=2$	probability charged after two transitions	probability low after two transitions

Because there are only two states, knowing p_t tells us the entire state distribution.

- A two-state policy reduces to one probability recurrence

Let $p_t = P(S_t = \text{charged})$ with $p_0 = 1$.

$$p_{t+1} = q(L) + [q(C) - q(L)]p_t$$

$q(C) = P(\text{charged next} \mid \text{charged now, chosen action})$

$q(L) = P(\text{charged next} \mid \text{low now, chosen action})$

$$\lambda = q(C) - q(L) \quad \text{and} \quad p^* = q(L) / [1 - q(C) + q(L)]$$

$$p_t = p^* + (p_0 - p^*)\lambda^t$$

$$V^{\pi} = \sum_t \gamma^t E[R_t \mid \pi]$$

This route evaluates each fixed policy from the charged start state.

■ The recurrence comes from the law of total probability

At time t , Nova is charged with probability p_t and low with probability $1-p_t$.

$$p_{t+1} = P(\text{C next} \mid \text{C now})P(\text{C now}) + P(\text{C next} \mid \text{L now})P(\text{L now})$$

$$= q(\text{C})p_t + q(\text{L})(1-p_t)$$

$$= q(\text{L}) + [q(\text{C})-q(\text{L})]p_t$$

First branch: charged now. Second branch: low now. Add the mutually exclusive ways to be charged next.

■ Solve the recurrence in two small steps

Write $\lambda = q(C) - q(L)$. A steady-state probability p^* stays unchanged after another transition. For $|\lambda| < 1$, deviations shrink; all four Nova policies satisfy this.

1 Find the fixed point

$$p^* = q(L) + \lambda p^* \Rightarrow p^* = q(L)/(1-\lambda)$$

2 Track distance from the fixed point

$$p_{t+1} - p^* = \lambda(p_t - p^*)$$

$$p_t - p^* = \lambda^t(p_0 - p^*)$$

$$\text{therefore } p_t = p^* + \lambda^t(p_0 - p^*)$$

■ EE policy: steady state plus a decaying transient

WORKED DERIVATION

EE chooses eco in both states: $q(C)=0.95$, $q(L)=0.50$, reward 7 when charged and 0 when low.

$$p_{t+1} = 0.50 + 0.45p_t$$

$$p^* = 10/11$$

$$p_t = 10/11 + (1/11)(0.45)^t$$

$$V^{EE} = 7[(10/11)/(1-\gamma) + (1/11)/(1-0.45\gamma)]$$

■ EE example: compute the first three probabilities by hand

EE uses eco in both states, so $p_{t+1}=0.50+0.45p_t$ and $p_0=1$.

Step	Substitute the previous probability	Charged probability
p_0	start charged with certainty	1.000000
p_1	$0.50+0.45(1)$	0.950000
p_2	$0.50+0.45(0.95)$	0.927500
p_3	$0.50+0.45(0.9275)$	0.917375

The probability falls quickly at first, then approaches $p^*=10/11\approx 0.9091$.

■ EE expected reward

Under EE, reward is 7 when charged and 0 when low. At time t : $P(\text{charged})=p_t$ and $P(\text{low})=1-p_t$.

$$E[R_t] = 7p_t + 0(1-p_t) = 7p_t$$

Time	p_t	Expected reward $7p_t$	Discounted contribution $\gamma^t \cdot 7p_t$
0	1.000000	7.000000	7.000000
1	0.950000	6.650000	$\gamma \cdot 6.650000$
2	0.927500	6.492500	$\gamma^2 \cdot 6.492500$
3	0.917375	6.421625	$\gamma^3 \cdot 6.421625$

■ EE example: the value is the sum of two geometric series

For EE, $p_t = 10/11 + (1/11)(0.45)^t$ and $E[R_t] = 7p_t$.

$$V^{EE} = \sum_{t=0}^{\infty} \gamma^t \cdot 7[10/11 + (1/11)(0.45)^t]$$

$$= 7[(10/11)\sum_{t=0}^{\infty} \gamma^t + (1/11)\sum_{t=0}^{\infty} (0.45\gamma)^t]$$

Use $\sum_{t=0}^{\infty} x^t = 1/(1-x)$ when $|x| < 1$.

$$= 7[(10/11)/(1-\gamma) + (1/11)/(1-0.45\gamma)]$$

■ The other three policies use the same recipe

FORMULAS FROM THE SAME RECURRENCE

$$\text{BB: } q(C)=.70, q(L)=.10, p^*=1/4, E[R_t]=2+8p_t \\ V^{\text{BB}} = 2/(1-\gamma) + 8[(1/4)/(1-\gamma) + (3/4)/(1-0.6\gamma)]$$

$$\text{EB: } q(C)=.95, q(L)=.10, p^*=2/3, E[R_t]=2+5p_t \\ V^{\text{EB}} = 2/(1-\gamma) + 5[(2/3)/(1-\gamma) + (1/3)/(1-0.85\gamma)]$$

$$\text{BE: } q(C)=.70, q(L)=.50, p^*=5/8, E[R_t]=10p_t \\ V^{\text{BE}} = 10[(5/8)/(1-\gamma) + (3/8)/(1-0.2\gamma)]$$

The first fraction is the long-run component. The second carries the charged-start transient.

Policy parameter summary for all four policies

ALL PARAMETERS SHOWN

For each policy: $\lambda = q(C) - q(L)$, $p^* = q(L) / [1 - q(C) + q(L)]$, and $E[R_t] = r(L) + [r(C) - r(L)]p_t$.

Policy	$q(C)$	$q(L)$	λ	p^*	Expected reward at t
EE	0.95	0.50	0.45	10/11	$7p_t$
BB	0.70	0.10	0.60	1/4	$2 + 8p_t$
EB	0.95	0.10	0.85	2/3	$2 + 5p_t$
BE	0.70	0.50	0.20	5/8	$10p_t$

Definitions: $p_t = P(S_t = \text{charged})$; $q(C) = P(\text{charged next} \mid \text{charged now})$; $q(L) = P(\text{charged next} \mid \text{low now})$, using the policy's action in that state.

Policy-value comparison worksheet

STUDENT WORKING

Y	EE	BB	EB	BE
0.6				
0.9				

At $\gamma=0.6$: best policy = _____
At $\gamma=0.9$: best policy = _____

Complete the table, then select the largest policy value in each row.

Official policy values: current Tutorial 3, p. 4.

■ Policy-value substitution at $\gamma=0.6$

BE has $V^{BE}=10[(5/8)/(1-\gamma)+(3/8)/(1-0.2\gamma)]$. Substitute $\gamma=0.6$ everywhere it appears.

$$V^{BE} = 10[(5/8)/(1-0.6) + (3/8)/(1-0.2 \cdot 0.6)]$$

$$= 10[(0.625/0.4) + (0.375/0.88)]$$

$$= 10(1.5625 + 0.426136...) = \underline{\hspace{2cm}}$$

Repeat the same substitution for EE, BB, and EB; then choose the largest value.

■ Part 1(b): when does the optimal policy change?

OFFICIAL QUESTION

Start charged. C/L=charged/low; E/B=eco/boost; policy XY uses X in C and Y in L. $P(C'|C, E/B) = .95/.70$; $P(C'|L, E/B) = .50/.10$; rewards C:E/B=7/10, L:E/B=0/2.

As the discount factor varies from 0 to 1, how does the optimal policy change? Provide an example discount factor for each distinct policy obtainable by varying γ .

Compare policy values at a small, medium, and very large γ .

Question: current Tutorial 3, p. 4.

■ Policy choices at three discount factors

STUDENT WORKING

Policy	$\gamma=0.20$	$\gamma=0.80$	$\gamma=0.98$
EE	_____	_____	_____
BB	_____	_____	_____
EB	_____	_____	_____
BE	_____	_____	_____

small γ
policy: _____

moderate γ
policy: _____

very large γ
policy: _____

Does every policy become optimal for some γ ? Justify your answer.

Examples and values: current Tutorial 3, p. 5.

■ Finding discount-factor switch points

STUDENT WORKING

A switch can occur only where two policy-value curves are equal. Denominators are positive for $0 \leq \gamma < 1$.

BB versus BE

$$V^{BB}(\gamma) = V^{BE}(\gamma)$$

Solve in $[0,1)$: $\gamma = \underline{\hspace{2cm}}$

BE versus EE

$$V^{BE}(\gamma) = V^{EE}(\gamma)$$

Solve in $[0,1)$: $\gamma = \underline{\hspace{2cm}}$

After finding the roots, test one γ in each interval and identify the largest policy value.

Policy-regime worksheet

STUDENT WORKING

Case or interval	Boundary / range	Best policy or tie
Starting point	$\gamma=0$	_____
First interval	$0<\gamma< \rule{1cm}{0.4pt}$	_____
First crossover	$\gamma= \rule{1cm}{0.4pt}$	_____
Middle interval	$\rule{1cm}{0.4pt} < \gamma < \rule{1cm}{0.4pt}$	_____
Second crossover	$\gamma= \rule{1cm}{0.4pt}$	_____
Final interval	$\rule{1cm}{0.4pt} < \gamma < 1$	_____

Use the policy-value formulas to fill every boundary and winning policy.

■ Part 1(c): can EB become optimal?

OFFICIAL QUESTION

Start charged. EB=eco when charged, boost when low. $P(C'|C, E/B) = .95/.70$; $P(C'|L, E/B) = .50/.10$; rewards $C:E/B = 7/10$, $L:E/B = 0/2$.

How could the MDP or discount factor be altered so that EB becomes optimal? Provide a modified MDP that requires minimal changes in probabilities, rewards, or discount factors.

Look for one parameter whose change weakens BE without weakening EB.

■ Policy-dependency worksheet

Choose one transition probability to change. Which policies use that state-action transition?

Policy	Action when low	Uses changed transition?	Direct effect
EE	eco	_____	_____
BB	boost	_____	_____
EB	boost	_____	_____
BE	eco	_____	_____

Predict which policies your proposed change strengthens, weakens, or leaves unchanged.

Minimal-change MDP worksheet

STUDENT WORKING

Proposed one-parameter change: _____

Policy	EE	BB	EB	BE
V at $\gamma=0.8$	_____ —	_____ —	_____ —	_____ —

At $\gamma=0.8$, recompute all four values and identify the largest: _____

Check which policies actually use your changed parameter before updating their formulas.

Official one-parameter modification and table: Tutorial 3, pp. 5-6. Formula typo corrected.

■ Modified-model recurrence worksheet

Let $x = P(\text{charged next} \mid \text{low now, eco})$. Choose x , then update only policies that use eco while low.

EE: $q(C)=0.95$, $q(L)=x$

$$p_{t+1} = x + (0.95 - x)p_t$$

$$p^* = x / [1 - 0.95 + x] = \underline{\hspace{2cm}}$$

BE: $q(C)=0.70$, $q(L)=x$

$$p_{t+1} = x + (0.70 - x)p_t$$

$$p^* = x / [1 - 0.70 + x] = \underline{\hspace{2cm}}$$

At $\gamma=0.8$: $V^{EE} = \underline{\hspace{1cm}}$, $V^{BE} = \underline{\hspace{1cm}}$. Compare them with the unchanged BB and EB values.

Modified-model value worksheet

Use your chosen x and $p_0=1$. Substitute each policy's p_t expression into $V=\sum_t \gamma^t E[r_t]$.

Always eco (EE)

$$p_t = p^* + (1-p^*)[0.95-x]^t$$

$$V^{EE} = 7[p^*/(1-\gamma) + (1-p^*)/\{1-\gamma(0.95-x)\}] = \underline{\hspace{2cm}}$$

Boost charged, eco low (BE)

$$p_t = p^* + (1-p^*)[0.70-x]^t$$

$$V^{BE} = 10[p^*/(1-\gamma) + (1-p^*)/\{1-\gamma(0.70-x)\}] = \underline{\hspace{2cm}}$$

Compare all four values at $\gamma=0.8$. Winning policy: _____

■ Part 1(d): practical meaning of γ

OFFICIAL QUESTION

What does the discount factor γ represent in practical terms? Why can $\gamma=1$ cause problems?

Interpretation
How fast does future value fade?

Mathematics
Can an infinite sum stay finite?

Question: current Tutorial 3, p. 6.

Discount-factor reasoning worksheet

STUDENT WORKING

future weight after k steps = γ^k

When $0 \leq \gamma < 1$ and $|R_t| \leq R_{\max}$

$|G| \leq \underline{\hspace{2cm}}$

When $\gamma = 1$

What can happen to the
infinite-horizon return on a
positive-reward cycle?

When can $\gamma = 1$ still give a well-defined return? Consider finite horizons and terminating tasks.

Effective-horizon worksheet

Effective planning horizon

rough horizon $\approx 1/(1-\gamma)$

$\gamma=0.6$: about _____ steps
 $\gamma=0.9$: about _____ steps
 $\gamma=0.98$: about _____ steps

Test $\gamma=1$ on a reward-1 loop

$1+1+1+\dots = \underline{\hspace{2cm}}$

What happens to this partial sum
as the horizon grows?

These are interpretations, not transition probabilities: stochasticity still comes from $P(s'|s,a)$.

■ Problem 1 Part 2: optimal policy setup

OFFICIAL SETUP

Nova's original two-state MDP. Discount factor $\gamma=0.8$. Value order: $[V(\text{charged}), V(\text{low})]$.

$$V^{(0)} = [0,0]$$

$$Q^{(0)}(c, \text{eco}) = Q^{(0)}(c, \text{boost}) = Q^{(0)}(l, \text{eco}) = Q^{(0)}(l, \text{boost}) = 0$$

One backup looks one step ahead from the current value estimates.

- $V^{(0)}=[0,0]$ is an initial guess, not a claim about Nova

Value order is [charged, low]. Superscript (k) counts algorithm iterations; subscript t counts time in the environment.

Environment time t

$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \dots$

Planning iteration k

$V^{(0)} \rightarrow V^{(1)} \rightarrow V^{(2)} \rightarrow \dots$

Starting from zero is convenient; Bellman backups progressively incorporate longer-term consequences.

■ Part 2(a-b): the first Bellman backup

OFFICIAL QUESTIONS

$\gamma=0.8$; C=charged, L=low; $V^{(0)}(C)=V^{(0)}(L)=0$. Rewards: $R(C,eco)=7$, $R(C,boost)=10$, $R(L,eco)=0$, $R(L,boost)=2$.

(a) What are the updated Q-values starting from $V^{(0)}$?

(b) What is the next updated value function $V^{(1)}$?

$$Q^{(1)}(s,a)=R(s,a)+\gamma \sum_{s'} P(s'|s,a) V^{(0)}(s')$$

Questions: current Tutorial 3, pp. 6-7.

■ Read a Bellman backup from the inside out

$$Q^{(k+1)}(s,a) = R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^{(k)}(s')$$

1

List successors

all next states s' that chance may produce

2

Average future value

$\sum P(s'|s,a) V^{(k)}(s')$

3

Discount

multiply that future average by γ

4

Add reward

add the immediate $R(s,a)$

■ First Bellman-backup worksheet

STUDENT WORKING

future term = $0.8 \times E[V^{(0)}(S')] = 0$

State	$Q^{(1)}(\text{eco})$	$Q^{(1)}(\text{boost})$	$V^{(1)} = \max Q^{(1)}$
charged	_____	_____	_____
low	_____	_____	_____

$V^{(1)} = [\text{____}, \text{____}]$

■ Part 2(c): VI and PI from the same $V^{(1)}$

OFFICIAL QUESTION

C/L=charged/low; E/B=eco/boost; $\gamma=.8$. Use your $V^{(1)}$ from Part 2(a-b), together with the supplied transitions and rewards.

Value iteration

Apply one Bellman optimality update to obtain $V^{(2)}$.

Policy iteration

Derive the greedy policy with respect to $V^{(1)}$, then evaluate it exactly.

Why should the two new value vectors differ after these operations?

Question: current Tutorial 3, p. 7.

Value-iteration worksheet

STUDENT WORKING

$\gamma=0.8$, $V^{(1)}(\text{charged})=10$, $V^{(1)}(\text{low})=2$.

Q from $V^{(1)}$	Calculation	Value
$Q(c, \text{eco})$	$7+0.8(0.95 \cdot 10+0.05 \cdot 2)$	_____
$Q(c, \text{boost})$	$10+0.8(0.70 \cdot 10+0.30 \cdot 2)$	_____
$Q(l, \text{eco})$	$0+0.8(0.50 \cdot 10+0.50 \cdot 2)$	_____
$Q(l, \text{boost})$	$2+0.8(0.10 \cdot 10+0.90 \cdot 2)$	_____

$V^2(\text{charged})=\max(_, _)=_ ; \text{action}=_$
 $V^2(\text{low})=\max(_, _)=_ ; \text{action}=_$

Policy-iteration worksheet

STUDENT WORKING

Your greedy policy from the preceding worksheet: $\pi(C)=$ _____ and $\pi(L)=$ _____.

$$V(C)=R(C,\pi(C))+0.8 \sum_s' P(s'|C,\pi(C))V(s')$$

$$V(L)=R(L,\pi(L))+0.8 \sum_s' P(s'|L,\pi(L))V(s')$$

Rearrange your two policy-evaluation equations into a linear system.

$$V^p(C)=\text{_____}, \quad V^p(L)=\text{_____}$$

■ Policy-evaluation algebra worksheet

Write your two rearranged policy-evaluation equations, then solve them simultaneously.

1 Isolate $V(L)$ from the second equation

Isolate one state value: $V(L)=$ _____

2 Substitute into the first equation

Substitute your expression for $V(L)$ into the first equation.

Solve for $V(C)=$ _____

Back-substitute to obtain $V(L)=$ _____

■ VI and PI comparison worksheet

VI backup: [_____,
_____]

One Bellman backup from $V^{(1)}$

Exact or approximate? Explain:

Policy _____: [_____,
_____]

What operation produced this vector?

Has optimality been established?

Explain why the two vectors can differ:

■ Part 2(d): what is the optimal policy?

OFFICIAL QUESTION

C/L=charged/low; E/B=eco/boost; $\gamma=.8$. Use your evaluated $V(C), V(L)$ from Part 2(c), together with the supplied transitions and rewards.

Use a one-step greedy check to determine the optimal action in each state.

$$\pi^*(s) = \arg \max_a \{R(s,a) + 0.8 \sum_{s'} P(s'|s,a) V(s')\}$$

A value vector is not itself a policy.

Question: current Tutorial 3, pp. 7-8.

■ Greedy-check worksheet

Use your evaluated $V(C)$ and $V(L)$. Compare eco and boost separately in each state.

$$Q(C,E)=7+.8[.95V(C)+.05V(L)]=\underline{\hspace{2cm}}$$

$$Q(C,B)=10+.8[.70V(C)+.30V(L)]=\underline{\hspace{2cm}}$$

$$Q(L,E)=0+.8[.50V(C)+.50V(L)]=\underline{\hspace{2cm}}$$

$$Q(L,B)=2+.8[.10V(C)+.90V(L)]=\underline{\hspace{2cm}}$$

charged action= $\underline{\hspace{2cm}}$ low action= $\underline{\hspace{2cm}}$

Does policy improvement change your policy? Justify using the comparisons.

■ Optimal-policy worksheet

STUDENT WORKING

State	Q(eco)	Q(boost)	Greedy action
charged	_____	_____	_____
low	_____	_____	_____

$\pi^*(\text{charged}) = \underline{\hspace{2cm}}$

$\pi^*(\text{low}) = \underline{\hspace{2cm}}$

Policy shorthand: _____. Optimal value: $V^* = [\underline{\hspace{2cm}}, \underline{\hspace{2cm}}]$.

Corrected answer and Q-values: current Tutorial 3, pp. 7-8.



03 Problem 2

Aster's Rescue Route: how four active RL methods use the same experience.

[QUESTION](#) [DISCUSSION](#) [REVEAL](#)

■ Problem 2: observed transitions from s_0 before the new episode

OFFICIAL SETUP

At s_0 : a=valley route, b=ridge shortcut. At s_1 : actions c and d. T is terminal. $\gamma=0.8$.

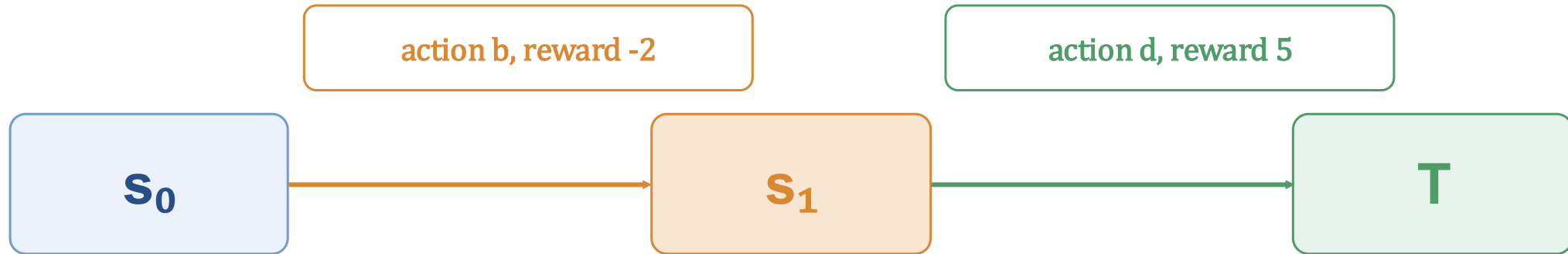
Action from s_0	Next state	Reward	Number observed
a	s_1	0	4
b	T	6	3
b	s_1	-2	1

The earlier b to s_1 episode continued with d, then reward 5 on entry to T.

■ The new episode and its return

NEW OBSERVATION

$S_0=s_0, A_0=b, R_1=-2, S_1=s_1, A_1=d, R_2=5, S_2=T$



$G \text{ after } (s_0, b) = -2 + 0.8(5) = \underline{\hspace{2cm}}$

The behavior policy actually selected d at s_1 .

■ Four update pipelines

Active ADP

LEARNS

What is learned?

USES EXPERIENCE

How is experience used?

Monte Carlo

LEARNS

What is learned?

USES EXPERIENCE

How is experience used?

SARSA

LEARNS

What is learned?

USES EXPERIENCE

How is experience used?

Q-learning

LEARNS

What is learned?

USES EXPERIENCE

How is experience used?

Fill both prompts for each method before attempting the detailed calculations.

■ Problem 2(a): Active ADP

OFFICIAL QUESTION

Counts after new data: $a \rightarrow s_1: 4$; $b \rightarrow T: 3$; $b \rightarrow s_1: 2$. Rewards: 0, 6, -2 respectively. $\gamma = 0.8$, $U(s_1) = 4$, $U(T) = 0$.

Update the estimated transition probabilities for actions a and b at s_0 . Then compute the expected value of each action and determine the greedy action.

$$P(s'|s_0, x) = N(s_0, x, s') / N(s_0, x)$$

$$Q(s_0, x) = \sum_s P(s'|s_0, x) [R(s_0, x, s') + 0.8U(s')]$$

Question: current Tutorial 3, p. 9.

Active ADP model-update worksheet

STUDENT WORKING

Action	Observed successor counts	Updated probabilities
a	$s_1: 4$	_____
b	T: 3, $s_1: 2$	_____

Use count ÷ total to update every probability. How did the new observation change the learned distribution?

For action a, compute $P(s_1|s_0,a)=count(s_0,a \rightarrow s_1)/count(s_0,a)=$ _____.

■ Active ADP action-value worksheet

STUDENT WORKING

Counts: $a \rightarrow s_1$ occurred 4 of 4 times; $b \rightarrow T$ occurred 3 times and $b \rightarrow s_1$ occurred 2 times. $\gamma=0.8$, $U(s_1)=4$, $U(T)=0$.

$$Q(s_0, a) = [4/4][0 + 0.8(4)] = \underline{\hspace{2cm}}$$

$$Q(s_0, b) = [3/(3+2)][6 + 0.8(0)] + [2/(3+2)][-2 + 0.8(4)] = \underline{\hspace{2cm}}$$

**Compare the two values. Greedy
action:**

Active ADP branch-value worksheet

Use the observed counts: $a \rightarrow s_1$: 4; $b \rightarrow T$: 3; $b \rightarrow s_1$: 2. Normalize the relevant counts to fill the probability column.

Chosen action	Chance outcome	Probability	Branch value $R + \gamma U$	Weighted contribution
a	s_1	_____	_____	_____
b	T	_____	_____	_____
b	s_1	_____	_____	_____

$Q(s_0, a) = \underline{\hspace{2cm}}$ $Q(s_0, b) = \underline{\hspace{2cm}}$ greedy action = $\underline{\hspace{2cm}}$

■ Problem 2(b): Monte Carlo control

OFFICIAL QUESTION

Before the new episode, complete returns after (s_0, b) were 6, 6, 6, 2. New episode: reward -2, then reward 5 one step later. $\gamma=0.8$.

Compute the return following (s_0, b) in the new episode and update the Monte Carlo estimate $Q(s_0, b)$ using all five returns.

Monte Carlo waits for the episode to finish before the full return is available.

Question: current Tutorial 3, p. 10.

■ Monte Carlo return worksheet

STUDENT WORKING

$$G = -2 + 0.8(5) = \underline{\hspace{2cm}}$$

returns: 6, 6, 6, 2,

$$Q(s_0, b) = (6+6+6+2+\underline{\hspace{1cm}})/5 = \underline{\hspace{2cm}}$$

Classify this update: model-based or model-free? bootstrap or complete return? when can it update?

■ Active ADP and Monte Carlo comparison

	Active ADP	Monte Carlo
Future information used	_____	_____
Detour branch / return	_____	_____
How outcomes combine	_____	_____
Result for (s_0, b)	_____	_____

Explain why the two estimators can differ after finite data:

■ Problem 2(c): SARSA and Q-learning

OFFICIAL QUESTION

$\alpha=0.5$, $\gamma=0.8$. Observe $(s_0, b, -2, s_1)$, then the behavior policy selects d at s_1 .

State	Action	Current Q-value
s_0	a	3
s_0	b	2
s_1	c	4
s_1	d	1

Compute the updated $Q(s_0, b)$ for SARSA and for Q-learning. Treat them as separate agents starting from this same table.

Question: current Tutorial 3, p. 11.

■ Anatomy of a temporal-difference update

$$\text{new } Q = \text{old } Q + \alpha(\text{target} - \text{old } Q)$$

current estimate

old $Q(s_0, b)=2$

target

**method chooses
the target**

TD error

target - 2

$\alpha=0.5$ means: move halfway from the old estimate toward the target

■ SARSA update worksheet

STUDENT WORKING

Old $Q(s_0, b) = 2$; $r = -2$; selected next action d ; $Q(s_1, d) = 1$; $\alpha = 0.5$; $\gamma = 0.8$.

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma Q(s', a') - Q(s, a)]$$

$$\text{target} = -2 + 0.8(1) = \underline{\hspace{2cm}}$$

$$Q(s_0, b) \leftarrow 2 + 0.5[\text{target} - 2] = \underline{\hspace{2cm}}$$

Which next action supplies SARSA's target, and whose behavior does that evaluate?

■ Q-learning update worksheet

STUDENT WORKING

Old $Q(s_0, b) = 2$; $r = -2$; next-state values $Q(s_1, c) = 4$ and $Q(s_1, d) = 1$; $\alpha = 0.5$; $\gamma = 0.8$.

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

$$\text{target} = -2 + 0.8 \max(4, 1) = \underline{\hspace{2cm}}$$

$$Q(s_0, b) \leftarrow 2 + 0.5[\text{target} - 2] = \underline{\hspace{2cm}}$$

Which next action supplies Q-learning's target? Must the behavior policy have selected it?

Reward-timing worksheet

Episode: $S_0=s_0$, $A_0=b$, $R_1=-2$, $S_1=s_1$, $A_1=d$, $R_2=5$, $S_2=T$. Online TD processes one transition at a time.

First transition

$(s_0, b, -2, s_1)$

Updates $Q(s_0, b)$; immediate reward is -2.

Second transition

$(s_1, d, 5, T)$

Separately: $Q(s_1, d) \leftarrow 1 + .5[5 + .8(0) - 1] =$

The question asks for the first-transition update using the supplied pre-update $Q(s_1, d)=1$.

■ Problem 2(d): what if the behavior policy selected c?

OFFICIAL QUESTION

Same transition $(s_0, b, -2, s_1)$, $\alpha=0.5$, $\gamma=0.8$, $Q(s_0, b)=2$, $Q(s_1, c)=4$, $Q(s_1, d)=1$. Now the behavior policy selects c.

What happens to the SARSA and Q-learning updates?

Compare the next action actually taken with the maximizing next action.

■ SARSA and Q-learning target worksheet

STUDENT WORKING

$Q(s_1, c) = 4$ and $Q(s_1, d) = 1$. Is selected action c greedy?

SARSA target = $-2 + 0.8Q(s_1, c) =$ _____

Q-learning target = $-2 + 0.8 \max_{a'} Q(s_1, a') =$ _____

Updated $Q(s_0, b) =$ _____. Do the two methods agree? _____

■ Problem 2(e): compare the four active RL methods

OFFICIAL QUESTION

Compare Active ADP, Monte Carlo control, SARSA, and Q-learning.

For each method, state:

what object it learns (a model or action values)

how the new experience enters its estimate

when the update can occur

how the improved estimate changes its policy

Question: current Tutorial 3, pp. 12-13.

Four-method comparison worksheet

STUDENT WORKING

Method	Model?	Learns	Target or estimate	Update timing
Active ADP				
Monte Carlo				
SARSA				
Q-learning				

Complete each row using the calculations and definitions from the preceding questions.

Comparison: current Tutorial 3, pp. 12-13.

■ Method-definition worksheet

Label	Precise meaning in this tutorial	Which methods?
model-based	_____	_____
model-free	_____	_____
bootstrap	_____	_____
on-policy	_____	_____
off-policy	_____	_____
active	_____	_____

Use the completed table to explain how Monte Carlo differs from the TD methods.

■ Problem 2(f): why exploration is necessary

OFFICIAL QUESTION

Why is exploration necessary for all four methods?

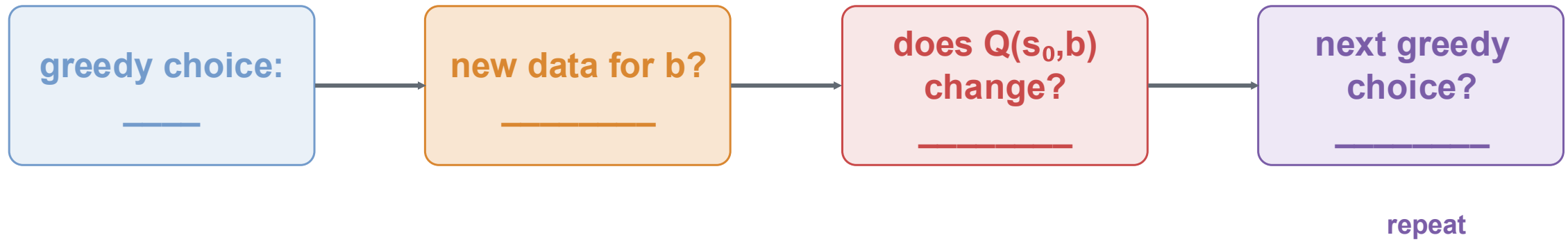
If the current favorite action is always chosen, what data remain missing?

What must ADP learn that the model-free methods do not?

Question: current Tutorial 3, p. 13.

■ Greedy-behavior discussion

Current estimates at s_0 : $Q(s_0, a) = 3$ and $Q(s_0, b) = 2$. A purely greedy behavior policy therefore selects a.



What might remain unknown if the behavior policy always selects the currently larger estimate?

■ Exploration discussion worksheet

STUDENT WORKING

Active ADP

What model-estimation problem occurs without trying different actions?

Model-free methods

What state-action coverage do value estimates require?

Propose an exploration rule: _____

How should exploration change over time? _____



04 Optional discussion

Reinforcement learning as a model of behavior, not a complete theory of people.

QUESTION **DISCUSSION** **REVEAL**

■ Optional Problem 3: reinforcement learning in the real world

OFFICIAL QUESTION

(a) Investigate how reinforcement learning ideas can be used to model human behavior and decision making.

(b) Choose one field: evolution and adaptive behavior, behavioral psychology, neuroscience, or education. Explain the connection to RL, discuss biologically grounded reward signals, and identify how insights from that field influenced RL methods in AI.

Name the mapping, then name where the mapping breaks.

Question: current Tutorial 3, p. 13.

Human-to-RL mapping worksheet

STUDENT WORKING

Human description	RL object	Example
current situation or belief	_____	_____
choice	_____	_____
evaluative feedback	_____	_____
expected future benefit	_____	_____
strategy for choosing	_____	_____
surprise relative to expectation	_____	_____

This is a model of selected mechanisms—not a claim that people optimize one known scalar reward.

■ Reward-prediction-error worksheet

Suppose $V(s_t)=5$, immediate reward $r_{t+1}=1$, next-state value $V(s_{t+1})=7$, and $\gamma=0.8$.

1 Form the one-step TD target

$$\text{target} = 1 + 0.8(7) = \underline{\hspace{2cm}}$$

2 Compare target with expectation

$$\delta_t = \text{target} - 5 = \underline{\hspace{2cm}}$$

Sign of δ_t	Meaning
positive	outcome is better than expected
negative	outcome is worse than expected
zero	outcome matches the current expectation

Dopamine activity can resemble prediction-error-like signals; dopamine is not simply pleasure or reward.



RL and AI discussion worksheet

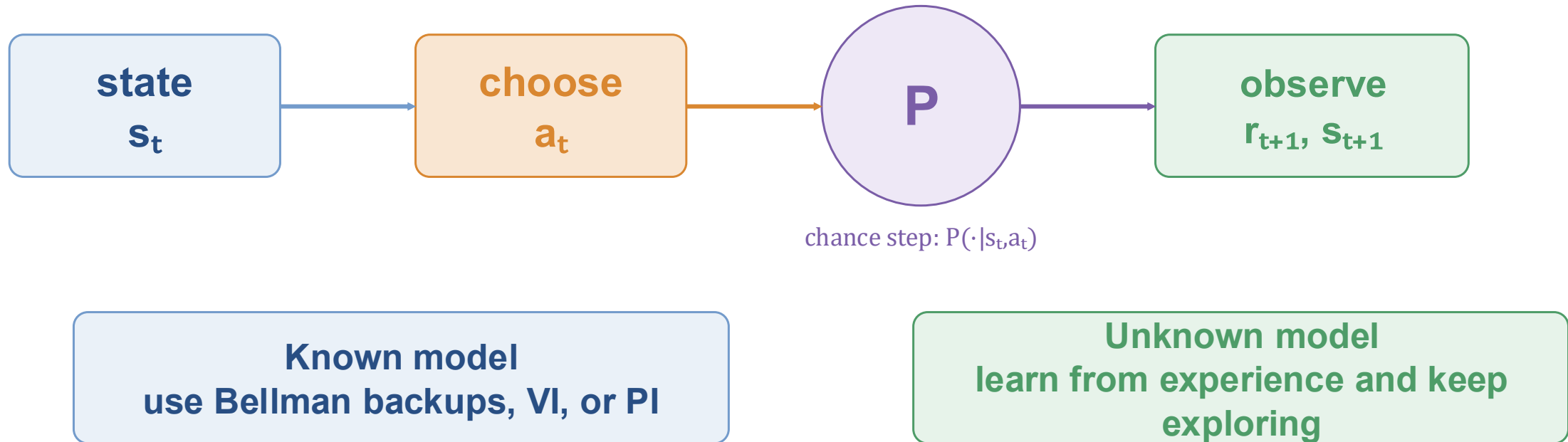
STUDENT WORKING

Field	RL connection	Influence on AI	Important limit
Behavioral psychology	_____	_____	_____
Neuroscience	_____	_____	_____
Evolution	_____	_____	_____
Education	_____	_____	_____

For each field, state one RL connection, one influence on AI, and one important limitation.

Discussion points: current Tutorial 3, pp. 14-15.

- One decision step connects planning and learning



Exit ticket: identify the choice, the chance event, and the learning target in one example.